MILITARY SPECIFICATION

# FRUIT BARS (Operational Ration Components)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers fruit bars for use by the Department of Defense as a component of Ration, Cold Weather, and the Food Packet Assault.

1.2 Classification. The fruit bars shall be of the following types as specified:

Type I - Fig
Type II - Blueberry

## 2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation.

SPECIFICATION

MILITARY

MIL-P-44276 - Packaging of Ration, Cold Weather Components (in Pouches)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A PSC 8920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists Inc., 3340 Pilot Knob Road, St. Paul, MN 55121.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

FIG ADVISORY BOARD

Administrative Rules and Regulations for the Marketing Order for Dried Figs, As Amended

(Application for copies should be addressed to the Fig Advisory Board, P.O. Box 709, Fresno, California 93712.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign materials, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Sugar, powdered or confectioner's. Powdered or confectioner's sugar shall be white and refined. The finished product shall contain not less than 96.5 percent sucrose, not more than 0.05 percent ash, not more than 0.80 percent moisture, and not less than 2.25 percent nor more than 3.25 percent of clean, edible starch thoroughly mixed, added to prevent caking. Not more than 2 percent shall remain on a US Standard No. 100 sieve and not less than 75 percent shall pass through a US Standard No. 200 sieve. The finished product shall not contain lumps larger than 1/2 inch in their greatest diameter that cannot be broken by light finger pressure.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

3.2.2 Shortening, vegetable. Vegetable shortening shall be refined, hydrogenated, deodorized vegetable oils and may contain antifoaming agents and antioxidants (active ingredients BHA, BHT, TBHQ, or propyl gallate) in amounts permitted by the FDA. The vegetable shortening shall possess a uniform plastic texture and shall have a stability of not less than 100 hours (A.O.M).

3.2.3 Salt. Salt shall be white refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.4 Whey protein concentrate, dry. The dry whey protein concentrate shall comply with the Food and Drug Administration Regulations for Direct Food Substances Affirmed as General Recognized as Safe (21 CFR 184.1979c). The concentrate shall have been spray dried using fresh pasteurized cheese whey. The product shall be free-flowing agglomerated and not more than 60 days old from time of manufacture in a plant approved by the Dairy Grading Section, USDA. The concentrate shall possess a pH value between 6.3 and 6.6 and shall possess a maximum moisture of 5.0 percent (see 6.3).

3.2.5 Invert syrup. Invert syrup shall be the aqueous solution of inverted or partially inverted sucrose. The syrup shall be colorless, odorless and flavorless except for sweetness. The syrup solids shall contain not more than 0.3 percent, by weight, of ash.

3.2.6 Corn syrup. Corn syrup shall be a clarified concentrated solution of the products obtained by incomplete hydrolysis of corn starch. The solids of corn syrup shall contain not less than 2 percent by weight of reducing sugars calculated as anhydrous dextrose.

3.2.7 Flour, wheat. Wheat flour shall be free-flowing, enriched, general purpose white (bleached).

3.2.8 Sodium bicarbonate. Sodium bicarbonate shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Calcium phosphate, monobasic. Monobasic calcium phosphate shall comply with the requirements of the Food Chemicals Codex.

3.2.10 Dried fig paste, Adriatic and Mission fig blend. Adriatic and Mission blend fig paste shall be a ground blend of dried Adriatic figs (80 percent) and dried Mission figs (20 percent). The fruit shall comply with the Federal Food, Drug and Cosmetic Act of 1938 and also with the current "Administrative Rules and Regulations for the Marketing Order for Dried Figs". Figs shall be ground through a grinder with a plate that has inside diameter holes of 5/64 inch standard with 7/64 inch maximum. Product shall contain no additives and be prepared from the latest seasons crop (see 6.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

3.2.11 Sugar, white, granulated. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof.

3.2.12 Citric acid, anhydrous, fine granulated. Fine granulated, anhydrous citric acid shall comply with the regulations of the Food Chemicals Codex.

3.2.13 Blueberry filling. Blueberry filling shall be an all natural fruit filling containing natural blueberries and shall possess distinct blueberry odor and flavor. Other ingredients may include sugar, corn syrup, dextrose, modified corn starch, evaporated apples, natural flavors, blueberry juice concentrate, pectin, citric acid, salt, vegetable gums, and caramel color. Sodium benzoate may be used as a preservative. Fruit content shall be a minimum of 17 percent. Brix shall be 77 ± 2 percent and pH 3.4 ± 0.2 percent. Brookfield viscosity shall be 40,000 centipoises ± 10,000 centipoises at 60°C at speed 20 while using spindle #7 (see 6.5).

3.2.14 Water. Water shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Preparation of fruit bar cookie jacket dough. The dough for the fruit bar cookie jacket shall be formulated and prepared as follows:

| Ingredient | Percent by weight |
|---|---|
| Flour, white, general purpose, enriched | 49.9 |
| Sugar, powdered or confectioner's | 14.8 |
| Water | 13.8 |
| Shortening, vegetable | 9.7 |
| Corn syrup | 6.9 |
| Invert sugar | 2.5 |
| Whey protein concentrate | 1.2 |
| Salt | 0.5 |
| Sodium bicarbonate | 0.5 |
| Calcium phosphate, monobasic | 0.2 |

a. Powdered sugar, vegetable shortening, salt, and whey protein concentrate shall be mixed in a Hobart or other suitable mixer until mixture is a finely divided crumb or just creamed.

b. Invert sugar and corn syrup shall be added to the crumb and mixed well until smooth.

c. Flour, sodium bicarbonate, and monobasic calcium phosphate shall be mixed together until homogeneous.

d. The water and flour mixture shall be added gradually to the sugar, shortening mixture until a smooth dough is formed.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

3.3.2 Preparation of fig bar filling. The fig bar filling shall be formulated and prepared as follows:

| Ingredient | Percent by weight |
|---|---|
| Fig paste, Adriatic and Mission fig blend | 61.0 |
| Sugar, granulated | 18.0 |
| Water | 8.8 |
| Corn syrup | 6.0 |
| Invert sugar | 6.0 |
| Citric acid, anhydrous, fine granular | 0.1 |
| Salt | 0.1 |

a. Mix fig paste, granulated sugar, citric acid, and salt in a Hobart or other suitable mixer until homogeneous.

b. Add the corn syrup and invert sugar to the paste mixture and continue mixing until homogeneous.

c. Add water and continue mixing until homogeneous.

3.3.3 Preparation of blueberry bar filling. The blueberry bar filling shall comply with the requirements of 3.2.13.

3.3.4 Preparation of fruit bar. The fruit bar shall be mechanically formed so that the finished fruit bar consists of not less than 50 percent fruit bar filling. Water percent adjustments for dough and paste may be made for proper machine forming such that finished product requirements in 3.5 are achieved.

3.3.5 Baking. The bars shall be baked so that the dough is cooked and is a golden brown color.

3.4 Pouch filling and sealing. Each pouch shall contain two fruit bars and be sealed in accordance with the requirements of 3.5 and MIL-P-44276 for Ration, Cold Weather.

3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The contents of each pouch shall weigh not less that 54.0 grams and not more than 67.0 grams.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

e. Moisture content shall not be greater than 20 percent.

f. Fruit bar cookie jacket shall be soft, but not soggy, and shall be of a golden brown color.

g. Fig fruit bar filling shall be smooth except for the fig seeds and of an odor and flavor typical of high quality fig bars.

h. Blueberry fruit bar filling shall be smooth except for the blueberry seeds and of an odor and flavor typical of blueberries.

i. Each bar shall measure 0.4 to 0.7 inches thick, 1.4 to 1.7 inches wide, and 3.1 to 3.5 inches long.

j. Fruit bars shall have cleanly and evenly cut ends and shall not be crushed or crumbled.

k. Fruit bars shall contain not less than 50 percent by weight of filling.

l. The filling in the fruit bars shall be evenly distributed.

3.5.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and "the plant sanitation requirements of the appropriate Government inspection agency."

3.7 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Examination for weight of filling. Fruit bars shall be examined to determine conformance to "filling" weight requirements. The lot size shall be expressed in pouches. The sample unit shall be the contents of seven pouches and the sample size shall be five. Pouches shall be randomly selected from the lot. Tare weight for jacket of fruit bars shall be established by weighing the sample units jackets and calculating the average tare weight applicable to one fruit bar. Lot shall be rejected if sample data indicate a lot average "filling" weight less than 50 percent of the total weight of the whole bars.



Source: https://assist.dla.mil -- Downloaded: 202610-22T01:10Z
Check the source to verify that this is the current version before use.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 54.0 grams or more than 67.0 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.5.5 Product inspection. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE I. Product defects 1/ 2/

| Category | | Defect |
|---|---|---|
| Major | Minor | |
| 151 | | Fruit bar cookie jacket hard or soggy and not a golden brown color - (Underbaked: Soggy; Overbaked: Burnt, excessively dark and hard) |
| 152 | | Fruit bar filling not smooth as specified. |
| 153 | | Fruit bar shape distorted. |
| 154 | | Fruit bar broken into more than three parts or has more than 1/4 of its corners/and pieces missing. |
| | 201 | Size not as specified. |
| | 202 | Flavor or odor not typical of fruit. |
| | 203 | Filling unevenly distributed. 3/ |

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by the Agricultural Marketing Service (AMS) agent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

3/ Unevenly distributed filling is filling that does not visually extend evenly throughout the interior of the jacket.

4.5.6 Moisture content testing. The moisture content of the fruit bars shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC): Chapter: Cereal Foods; Section: Fig Bars and Raisin Filled Crackers; Method: Vacuum Oven Method. Any result not conforming to the moisture content requirement in 3.5e shall be scored as a major defect. The lot size shall be expressed in units of pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44276 for the Ration, Cold Weather.

5. PACKAGING

5.1 Packaging. Packaging of the Fruit Bars shall be in accordance with the applicable requirements of MIL-P-44276.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of product required (see 1.2).
c. When a first article is required (see 3.1, 4.4 and 6.2).
d. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article inspection. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Whey protein concentrate. The whey protein concentrate used in in-house production was Vari-Dairy Plus N manufactured by Nutrisearch International Corporation. Other whey protein concentrate meeting the requirements of 3.2.4 may be equally acceptable.

6.4 Dried fig paste, Adriatic and Mission fig blend. The fig paste used in in-house productions was Blue Ribbon Fig Paste #00619 80:20 Adriatic:Mission Figs manufactured by Sun Diamond Growers. Other fig pastes meeting requirements of 3.2.10 may be equally acceptable.

6.5 Blueberry filling. The blueberry filling used in in-house production was Natural Blueberry Filling WL-11785 manufactured by Florasynth, Inc. Other blueberry filling meeting the requirements of 3.2.13 may be equally acceptable.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

6.6 Subject term (key word) listing.

Blueberry bars
Cookie bars
Fig bars
Fruit bars

Custodians:

Army - GL

Review activities:

Army - MD, TS
Navy - SA, MC, MS
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 8920-0494



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-F-44279(GL) | Fruit Bars (Operational Ration Components) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 63 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:10Z
Check the source to verify that this is the current version before use.